Shruti Swaminathan

5th year BS-MS (Biology) student at IISER Mohali, India
MS Thesis at the Friedrich Miescher Laboratory (MPG), Tübingen
DST-INSPIRE Scholar

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Education

Indian Institute of Science Education and Research Mohali — **2022 – 2027**
Integrated BS-MS Degree – ***Current CPI: 9.74/10*** — *Mohali, India*
A 5-year bachelor's and master's degree with a **major in Biological Sciences and a minor in Data Science**

Experience

Diffusiophoresis in self-organizing systems *in vitro* — **June 2026 – present**
Masters Thesis | Friedrich Miescher Laboratory of the Max Planck Society — *Tübingen, Germany*

- Guide: Dr. Beatrice Ramm
- Selected under the IISER-MPG program to conduct the research for my Masters thesis at the Max Planck Institute
- Looking at the effects of diffusiophoretic transport in the two self-organizing protein systems in vitro and potential coupling of the systems via diffusiophoresis and further go onto to model these interactions
- Protein purification using Ni-NTA chromatography, preparation of single unilamellar vesicles (SUVs) and 2D supported lipid bilayers (SLBs) of varying lipid compositions, *in-vitro* membrane reconstitution assays, confocal microscopy, image analysis via Fiji macros and Python pipelines

Microbial evolution through the lens of phages — **May 2025 – July 2025**
Summer Internship | IISER Thiruvananthapuram — *Trivandrum, India*

- Guide: Dr. Yashraj Chavhan
- Investigated bacteriophage–bacteria interactions focusing on the role of bacterial multicellularity in phage resistance, phage isolation and characterisation
- Quantified bacterial growth and co-culture dynamics using growth curves and CFU counts; modelling of microbial population dynamics.

Tracking Behaviour of the Asian Elephant Population in the Kabini backwaters — **May – July 2024**
Summer Internship | Jawaharlal Nehru Centre for Advanced Scientific Research (JNCASR) — *Bengaluru, India*

- Guide: Prof. TNC Vidya
- Lagged associations in adult non-musth male elephants: Developed MATLAB (MATLAB version R204a) code to spatially cluster observations across temporal intervals and conducted social network analysis to investigate patterns of male-male associations in non-musth Asian elephants.
- Land-use mapping: Analysed temporal changes in the Kabini study area using Landsat-8 imagery, QGIS, and Google Earth; generated a Land Use/Land Cover (LULC) map using supervised classification with the QGIS Semi-Automatic Classification Plugin.

Understanding the role of *Drosophila melanogaster* in developmental biology — **May – August 2023**
Summer Internship | IISER Mohali — *Mohali, India*

- Guide: Prof. Lolitika Mandal
- Fly management from larval to pupal stage, fly crosses, balancer chromosome study, imaginal disc dissection and mounting, nuclear staining (Hoechst and DAPI), spatial regulation using Dpp Gal4 system with various reporters like GFP and eyeless under regulation of UAS, temporal regulation using Gal80^{ts}, Cell lineage tracing and analysis using G-TRACE.

Workshops and Conferences

Ulm Biophysics Symposium 2026: Quantifying Biology Across Scales — **October 2026**
Biophysics Symposium | Ulm University (presented a poster) — *Ulm, Germany*

6th VAAM Discussion Meeting: Microbial Cell Biology — **October 2026**
Conference | Marburg (presented a poster) — *Marburg, Germany*

Spins, Games and Networks: From Data to Discovery in Complex systems — **December 2025**
Workshop on Complex Systems | Institute of Mathematical Sciences (IMSc), Chennai — *Chennai, India*

Unifying Theories in High-Dimensional Biophysics — **July–Aug 2025**
Conference | International Centre for Theoretical Sciences (ICTS – TIFR) — *Bengaluru, India*

Science Media Workshop — **March 2025**
Science Communication Workshop — *IISER Mohali*

Other Projects

Modelling a Tale of Two Species: ODEs in Microbiome Research **2025**

IDC402: Nonlinear Dynamics and Chaos

Term paper for the Nonlinear Dynamics coursework: Explored the Lotka-Volterra and consumer-resource models to reproduce results from *Piccardi et al.'s* (2019) work on bacterial coculture dynamics

Like a Rolling Stone: Exploring Waddington's Landscape in the Context of Cell Fate Decisions **2025**

IDC401: Theoretical Biology

Term presentation for the Theoretical Biology coursework: Explored Waddington's ideas in the context of cell fate decisions using geometry and dynamical systems theory. Reproduced results from existing literature on mathematical modelling of Waddington's genetic assimilation experiments landscapes *(Raju et al. 2023)*.

MEGA Plate under the supervision of Prof. N.G. Prasad & Dr. Shravan Mishra **September 2023**

***M**icrobial **E**volution **G**rowth **A**rena* *IISER Mohali*

Our aim was to standardise the MEGA Plate Setup for demonstrating evolution on an ampicillin gradient. Referenced and modified from the paper "Spatiotemporal microbial evolution on antibiotic landscapes" by Dr. Michael Baym et al.; used *Escherichia coli DH5α transformed with pET-29b(+) plasmid with* kan^R *cassette*

Awards and Scholarships

Academic Merit Certificate **2024-2026**

*Awarded **5** times by the institute for the best academic performance during the* 4th *to* 8th *semesters*

Zonal Topper Award - Mimamsa, IISER Pune **January, 2025**

Part of the Zonal Topper winning team for the states of Punjab and Haryana in the national-level annual science competition - Mimamsa, organized by IISER Pune for students at the undergraduate level

Summer Research Fellowship Program **April 2024**

Awarded by the Indian Science Academies: Indian Academy of Sciences - Indian National Science Academy - National Academy of Sciences, India (IASc-INSA-NASI); (Did not accept the fellowship)

DST INSPIRE Scholarship **2022-2027**

Scholarship for Higher Education (SHE) under the Department of Science and Technology, Government of India; awarded for having aggregate marks within the top to the top 1% of the Class XII national board examination

Extracurricular Activities

Tutor- Genetics and evolution **Spring semester 2025**

Conducting regular help sessions for junior batches for their Genetics and evolution course *IISER Mohali*

Tutor- Genes and development **Monsoon semester 2024**

Conducting regular help sessions for junior batches for their Genes and development course *IISER Mohali*

Biology Department Representative, Member of SRC **2024-2025**

Member of the Student Representative Council (SRC) *IISER Mohali*

Core Organizing Committee, IISER Mohali's 18th Foundation Day **September 2024**

Organized a science outreach event at IISER Mohali with scientific demonstrations, lectures, quizzes and competitions for 2000 school students across the tri-city *IISER Mohali*

Co-Convenor, Biology Discussion Forum (BDF) **2023–2024**

IISER Mohali's Biology Club *IISER Mohali*

Member, IISER Mohali's Sports Contingent **2023–present**

Representing IISER Mohali in the Inter IISER Sports Meet as part of the Basketball, Football and Athletics Team *IISER Mohali*

Member, Manthan's Outreach Team **2023– present**

IISER Mohali's student-run magazine *IISER Mohali*